Knowing Is Not Committing

Replicating MIT's weight-space edit and programming a truth-gated reflex into a 0.8B reasoning model · interim report

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Code, data and pre-registration log: github.com/NeoCortex0815/knowing-is-not-committing · questions and corrections are welcome as GitHub issues.

> *"Find out whether the model has the solution, and when, and let the inner push through, despite the pressure."*
> — Dave Uhlemann, 6 Nov 2026

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> **Status of the evidence.** We mark every claim as one of three kinds:
> - **[pre-registered]**: tested with a reading fixed in a dated log before the run;
> - **[descriptive]**: measured, but not pre-registered;
> - **[interpretation]**: our reading of the data.
>
> All results come from a 0.8B model.
> - Most results rest on a single seed; second seeds exist for parts of the data.
> - Part of the data comes from a development set that we used several times.
> - MATH-500 problems 161–300 served as the confirmation set (Section 3.9).
> - Problems 301–370 were used once, late on 8 Nov, for a second confirmation of the answer-mode results, with readings fixed in advance (Section 3.10). Problems 371–500 remain untouched.
>
> **What we mean by "knowing".** A frozen linear probe on the model's hidden states rates the correct value as true while it is in the context. This is a measurement, not a claim about certainty.

Abstract. MIT's weight-space edit reduces reasoning loops in Qwen3.5-0.8B. We rebuilt that edit independently, wrote a truth-gated reflex into 16 MLP neurons, and ran a series of pre-registered tests on fresh MATH-500 problems.

- Three findings carry the report:
  - (1) The model often *knows* a correct answer long before it *commits* to it. A linear probe reads the truth inside the reasoning. Our reflex closes the reasoning at the right moment, but a close imposed from outside is not a commitment: on fresh problems the model reopens its reasoning after 77 % of its closes.
  - (2) Without its thinking block (answer mode), our edited model is close to MIT's edit with thinking: 71 vs 80 of 140, at less than half the tokens per correct answer.
  - (3) When two independent answer-mode runs agree, 91 % of the answers are right (pooled over three sets).
- Sampling settings alone move MIT's own edit from 25 % to 55 %.
- We report every refuted reading alongside the confirmed ones and pre-register the next test publicly.

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TL;DR

1. **Sampling protocol decides half of the effect.** On the same 40 MATH-500 problems, MIT's published edit scores **55 %** with vLLM-style sampling and **25 %** with llama.cpp's default sampling. [descriptive]
   - The llama.cpp default differs in two ways: the repetition penalty only looks back over the last 64 tokens, and temperature is applied last.
2. **The loop direction is findable without a search agent.** Our own direction (cosine 0.76/0.77 to MIT's) lifts the base model from **10 % to 37.5 %** on those 40 problems [descriptive]. MIT's edit reaches 55 % there.
   - We used the same edit recipe and derived the direction from base-model transcripts.
3. **ERGORE ties MIT on problems 1–160, but not on fresh problems.** ERGORE is our own loop direction plus an explicit, conditioned *closing* push. [pre-registered]
   - 1–160: **79 vs 77** (p = 0.85). ERGORE uses 13 % fewer tokens.
   - **Fresh problems 161–300: 65 vs 80 of 140** (7 : 22, p = 0.008; −10.7 points, 95 % CI −18.0 to −3.4). The reading fixed in advance is "MIT level not reached".
   - A second seed on one block (261–300) shrinks that block's gap from 14 : 22 to 16 : 16. [descriptive] The gap is real, but probably smaller than seed 0 suggests.
4. **Models know more than they act on.** [pre-registered / descriptive]
   - A linear truth probe at layer 13, upstream of every edit, separates correct from wrong candidate answers inside the reasoning (AUC 0.96 on practice data). In the field it is conservative: on the test's final answers it flags 1 of 86 correct answers as unsure and catches 8 of 16 wrong ones.
   - The model keeps thinking for a median of **~2,000 tokens (T = 0.6) to ~6,000 tokens (T = 1.0) after it first writes the correct answer**.
   - Forcing a final answer at the end of the budget lifts every model: one base-like variant goes from 11 to 28 of 60. It also shrinks the differences between edits to non-significance.
5. **Blunt pressure has a price.** [pre-registered as a harm check / descriptive]
   - Our unconditional closing push costs instruction following: **IFEval 61/120**, against 74 without the push (p = 0.041) and 87 for MIT.
   - It also produces "empty commits", where the model stops thinking without a solution. These are 22 of 26 errors on a development set.
6. **A truth-gated trigger written into the weights (WT-A).** Its mechanism works on fresh problems; its accuracy does not reach MIT's.
   - WT consists of 16 neurons in layer 15. They fire only when a candidate answer has just been written *and* the internal truth signal is high. They then push along the measured closing gradient.
   - **Development set** (n = 60): 31 vs 23 correct against the same model without the trigger, and +8 again with a second seed [descriptive]. 64 % fewer tokens on problems both solve. The pre-registered gate stayed closed because a probe check failed.
   - **Fresh problems 161–300**, test fixed before any of these runs existed (H-WT2): WT-A closes right after the correct value in **65 %** of opportunities, MIT in **36 %** (p = 0.0001). Its fast closes are right in 61 % vs 44 %. [pre-registered]
   - **Accuracy there: 61 vs 80 of 140** (−13.6 points, paired bootstrap 95 % CI −21.4 to −5.7). [descriptive] With a second seed on 161–260, 47 vs 58 of 100.
7. **Why the trigger is not enough** [descriptive, post hoc]:

- **One path to success:** 51 of WT-A's 61 correct answers follow closes driven by the trigger (Section 3.9). MIT also checks for ~12,000 more tokens and then commits on its own (44 of 80).
- **A forced close is not an inner close:** after a trigger close, WT-A reopens its reasoning in the answer part (a second `</think>`) in 83–100 % of development runs. On fresh problems the figure is 77 % of all closes and 58 of 61 correct runs, against 16 % for MIT.
  - Among the 32 silent fresh runs, which never deliver, the rate is 12 of 32. The three numbers describe different groups of runs.
  - WT-A's successful path is almost always: trigger close, renewed reasoning, then its *own* second close, then delivery.
- **Depth:** the truth signal fades with *loops and difficulty*, not with depth into the reasoning.

8. **An honest "I don't know" from the inside signal.** [pre-registered] The frozen truth probe, read at the delivered answer, separates forced answers with AUC 0.83.
   - Abstaining when it stays low turns forced answering from harmful into useful: score (correct − 2·wrong)/N goes from −0.30 to +0.17.
   - 327 of 380 abstentions would have been wrong.
9. **Answer mode ("front room") is our strongest own path.** Without the thinking block, the model reasons visibly in its answer.
   - Development range [descriptive]: our own-direction model solves 77 of 160, exactly as many as MIT with thinking (22 : 22), at ~40 % of the tokens.
   - Fresh problems 161–300 [pre-registered]: **71 vs 80 of 140** for MIT with thinking (−6.4 points, 95 % CI −15.0 to +2.1). That is "open" by the reading fixed in advance: holds at ≥ 73, refuted at ≤ 66. It needs 16,600 instead of 37,700 tokens per correct answer.
   - MIT itself in answer mode reaches only **59 of 140** (27 : 6 against MIT with thinking, p = 0.0003) [pre-registered]. A back room pays off when it has a working exit; ours does not have one yet.
   - The advantage is not an artefact of the presence penalty: the same +2 with penalty 0 and 1.5 (n = 20) [pre-registered]. The front room is not more stable across seeds than the back room: 75 % of problems are solved the same way in both seeds [pre-registered, refuted].
10. **The front room knows when it is sure: a council of intuitions.** Several answer-mode runs answer independently. When they agree, the model delivers; only disputed cases go to the back room.
    - **Two intuitions** [pre-registered, three confirmations]. When two runs with different seeds give the same answer, it is right in **37 of 38** cases on 161–260, 13 of 17 on 40 new training problems, and **28 of 31 on the untouched problems 301–370**. Pooled over the three sets, 78 of 86 are right (91 %); the pooled figure is descriptive, because each set was pre-registered on its own. When the runs disagree, the first run is right in only 13–32 %.
    - **Three intuitions** [pre-registered, "open" by one point]. A majority of three delivers on 62 of 100 problems instead of 38, and 89 % of these are right (target: 90 %). The net score (correct − 2·wrong)/N rises from +0.35 to +0.41. When all three agree, 29 of 29 are right.
    - **A probe trained on answer text** [pre-registered; "open" in all three tests]. It reads correctness at the delivered answer with AUC 0.76–0.85 on new data; the probe trained on thinking text reaches 0.68.
      - It adds only +0.01 to +0.05 over answer length and difficulty.
      - Its apparent edge on quick answers did not replicate on untouched problems (0.61 vs 0.60).
      - Used as an extra delivery gate it hurts: score +0.13 instead of +0.31 with agreement alone.
      - Used as a **referee** between two disagreeing intuitions it helps [pre-registered]: 11 instead of 7 correct answers. The agreeing cases are not touched. With only 18 decisive problems this is fragile: the bootstrap 95 % interval for the gain is −1 to +10.
    - **Switching rooms** [pre-registered / descriptive].

    - With two intuitions, falling back to WT-A's back room loses: 54 vs 57 of 100 for answer mode alone [pre-registered].
    - With three intuitions it wins: 62 vs 57 [descriptive].
    - With MIT's back room as fallback: 63 (two intuitions) and 66 (three), against 58 for MIT alone [descriptive].
    - On the untouched problems 301–370 [pre-registered]: "two agree → answer mode, otherwise MIT's back room" solves **40 of 70**. MIT alone solves 38 and answer mode alone 35. Agreement precision is 90 %, but the rule costs more tokens per correct answer than MIT (56,800 vs 41,000). The reading is therefore **open**. The gain comes from both rooms: precision in front (28 vs 26 on the agreeing problems) and MIT's exit at the back (12 vs 7 on the rest).
  - **Two modes, two costs.**
    - A *single* answer-mode run is the efficient path: 16,600 instead of 37,700 tokens per correct answer (item 9).
    - Asking *several* intuitions buys precision with tokens. Combined with a back room, it can cost more per correct answer than MIT alone.
11. **Negative results.** [pre-registered]
    - A first gate along a state-difference direction failed all criteria.
    - A "thinking switch" based on the inner signal at the end of the prompt was refuted (p = 0.13).
    - Probe cascade, notes for a second chance and loop-based "state doors" were refuted (Section 3.10).
    - Inserting `</think>` from outside does not produce an inner "finished", not even at the model's own closing phrase. MIT then delivers in 36 % of cases, against 18 % without a closing phrase [descriptive].
    - Replaying WT-A's 32 silent answer parts rescues almost nothing.
      - We re-ran it with a fair cap: our first cap of 1,024 tokens was too short, because even WT-A's own correct answer parts have a median of ~2,000 tokens.
      - With 4,096 tokens, 0 of 14 runs are rescued when a second `</think>` is banned, and 2 without the ban.
      - The two rescues came after the model closed its renewed reasoning itself.

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1. Background

> *"Is the model simply unable to say 'I don't know'?"* — D. U., 7 Nov 2026
>
> This question led to the answer-forcing probe (Section 3.4) and to the honesty track (Sections 3.8 and 6).

On 2026-11-02, MIT CSAIL published *"Sparks of Weight-Space Programming"* (Takehi, Teoh, Zhang, Khattab).

- **What it shows:** in their Experiment 8, a general LLM edited Qwen3.5-0.8B directly in weight space.
- **The edit:** subtract one "loop" direction from 64 output columns of `mlp.down_proj` in layers 15 and 19, about 0.016 % of the weights.
- **The effect:** MATH-500 rose from 7.9 % to 48.1 %. The authors read this as *removing an obstacle that hid an ability*, not as adding one.

Our questions:

- Can the edit be reproduced and understood on consumer hardware?

- Can the "stop looping, start answering" behaviour be decomposed into measurable parts?
- Can it be steered more precisely, and at what cost?

Closest prior work (position explicitly):

- Zhang et al. (2025) show that reasoning models encode the correctness of intermediate answers in their hidden states. An *external* probe then supports early exit (up to 24 % fewer tokens at equal accuracy).
- Orgad et al. (2025) show the same "knows more than it shows" gap for hallucinations.
- More broadly, readable signals about the correctness of what a model says are well established (Kadavath et al., 2022; Azaria and Mitchell, 2023; Ferrando et al., 2024). Single directions in activation space can mediate whole behaviours (Arditi et al., 2024); MIT's edit and ours are of this kind.
- Our contribution is not the gap itself:
  - We dissect a published weight-space edit on consumer hardware. The sampling protocol accounts for half of its effect, and the loop direction can be found without a search agent.
  - We show that the bottleneck is committing, not knowing (truth probe, answer forcing).
  - We write a truth-gated exit *into the weights*, with no external controller. Its mechanism is confirmed on fresh problems; its accuracy does not reach MIT's.
  - We measure *inside* why: a teacher-forced trigger audit (Section 3.9).
  - We report two failure modes of hand-written reflexes: the negative SiLU lobe, and a phase failure, where a forced close is not an inner close.
  - We show that the same frozen probe supports an honest "I don't know".
  - We report our strongest edit's failure to generalise: ERGORE matched MIT on 1–160, but not on fresh problems.

2. Setup

- **Model:** Qwen3.5-0.8B (hybrid Gated-DeltaNet / full attention, 24 layers), thinking mode.
- **Hardware:** one laptop GPU (8 GB) via llama.cpp (q8_0 GGUF). A second machine (CPU and iGPU) runs further evaluation blocks with identical model files (checked by checksum) and identical settings.
- **Protocol** (as in MIT's evaluation):
  - temperature 0.6, top-p 0.95, top-k 20, presence penalty 1.5;
  - up to 32,768 new tokens, seed 0;
  - the last `\boxed{}` is graded with math-verify.
- **vLLM-style sampling in llama.cpp:** repetition penalty over the whole answer; sampler order penalties → temperature → top-k → top-p → min-p.
- **Problem sets:**
  - MATH-500 problems 1–160 as test set;
  - MATH-500 161–300 as confirmation set; 301–370 as a second, late confirmation; 371–500 untouched;
  - MATH train problems as practice set;
  - "Set B": 60 fresh MATH train problems with the MATH-500 level mix, used as development set;
  - IFEval: 120 prompts, strict prompt-level scoring.
- **Pre-registration:** before each decisive run, the reading (criteria, thresholds) is written into a dated log. Later changes are logged with their reason, and only before the corresponding data exist.
- **Statistics:**

- Most pre-registered readings are threshold rules ("holds / open / refuted"). Where a test also has a p-value (exact McNemar for paired accuracy, Fisher for rates), we report it unadjusted.
- The three main effects with p-values remain significant after Holm correction over all ~30 pre-registered readings of this project:
  - the trigger's closes after the correct value (H-WT2), $p = 0.0001$;
  - MIT's back room against its front room, $p = 0.0003$;
  - agreement of two intuitions, $p \approx 10^{-11}$.
- Descriptive results carry no p-value as evidence; we give counts and effect sizes.

3. Results

3.1 Sampling decides half of the effect [descriptive]

Same 40 MATH-500 problems, MIT's published edit:

Setting	correct	stopped at token limit
MIT's reported run on these 40	47.5 %	~33 %
our run, vLLM-style sampling	**55.0 %**	30 %
our run, llama.cpp default sampling	**25.0 %**	67.5 %

- **Base model, vLLM-style:** 10 % correct, 92.5 % truncated.
- **Base → MIT edit:** 10 % → 55 %, with 20 problems rescued and 0 lost.

Practical note: anyone running repaired models in llama.cpp should set `repeat_last_n` to the full answer length and put temperature before the filters.

3.2 Our own loop direction [descriptive]

- **Construction:** the direction is the mean difference "repetitive minus productive" over base-model transcripts of 48 practice problems. The recipe is otherwise MIT's: layers 15 and 19, top-64 columns, α = 6.
- **Overlap with MIT:** cosine 0.76 and 0.77 to MIT's direction.
- **Result on the 40 problems:** 10 % → **37.5 %**, truncations 92.5 % → 60 %. Every problem ours solves, MIT's edit also solves. [interpretation] Same mechanism, weaker dose.
- **Knife-edge finder.** From points shortly before a loop starts, we let the base model branch 6–8 times (256 tokens each) and looked for "knife edges": points where some branches loop and others stay productive. We found 31 knife edges.
  - The share of looping branches grows the later the branch point lies (11 → 13 → 28 → 40 branches).
  - [interpretation] In the base model, loops build up gradually rather than tipping at a single point.
  - A direction built from the knife edges alone was noisy (cosine 0.77 to MIT's, low stability). Mixed with our direction it reached 0.79–0.80.

3.3 ERGORE: own direction plus an explicit closing coupling [pre-registered]

Construction. On layer 15 we add a second rank-1 term to the inversion. When the "loop detector" columns fire, it pushes along our own *commit axis*: "16 tokens before `</think>`" minus "middle of thinking". Its trigger is a whitened ("sharpened") detector. Layer 19 keeps the inversion only.

MATH-500	ERGORE	MIT	only ERGORE : only MIT	McNemar p	mean tokens ERGORE / MIT
1–80	43/80	43/80	4 : 4	1.0	18,557 / 21,501
81–160 (fresh)	36/80	34/80	10 : 8	0.81	18,489 / 21,248
1–160	**79/160 (49.4 %)**	**77/160 (48.1 %)**	14 : 12	0.85	18,523 / 21,374

- **Reading, fixed in advance:** "MIT level reached", on both halves. Not "above MIT": the 95 % CI of the difference is −5.0 to +7.5 points.
- **Tokens:** 13.3 % fewer overall (88 vs 55 problems cheaper, p = 0.007). There is no saving on problems both solve. The saving comes from problems neither solves, where ERGORE stops earlier.

3.4 Models know more than they act on

Truth probe. [pre-registered on test answers]

- **Construction:** logistic probe on layer-13 activations at the end of each candidate answer in the reasoning. Trained on practice data only, then frozen.
- **Practice data:** AUC 0.96.
- **Test final answers, applied once:** it flags **1 of 86** correct answers as unsure and catches **8 of 16** wrong ones.
- **Development set B:** AUC 0.925 on ERGORE runs and 0.966 on MIT runs. [descriptive]

Hearing the bell is not enough. [descriptive]

- **Setup:** 20 easy practice problems, our own-direction edit. The prompt adds "think for at most about 1,500 words, then give your final answer". Budget 6,144 tokens.
- **Result:** with this soft announcement, **0 of 20** runs finished within the budget (1 of 20 without it).
- **What the texts show: 10 of 20** thinking texts write the constraint down themselves ("Constraints: … max ~1500 words …") and keep thinking anyway.
- [interpretation] The model does not lack the information that it should stop; it lacks the drive to stop. A soft instruction does not help. What helps is an inner push (a weight edit) or a hard external stop.

The signal is a flash. [descriptive] It peaks at the candidate end (AUC 0.96) and falls within a few tokens (0.67–0.80 at +8 to +200 tokens). The model closes its reasoning on average ~600 tokens after the last candidate.

Late closing. [descriptive, Set B] Tokens between the *first correct* candidate and `</think>` on solved problems (median):

run	median tokens
ERGORE, T = 0.6	2,024
ERGORE, T = 1.0	6,073
MIT, T = 1.0	6,886

Answer forcing. [descriptive] Like budget forcing in s1, we append "The final answer is \boxed{" to every run that ended without a clean answer, identically for all models. Set B, 60 problems:

	without forcing	with forcing
ERGORE, T = 0.6	34	38
ERGORE, T = 1.0	21	29
MIT, T = 1.0	24	32
base-like variant ("Gatter"), T = 0.6	27	36
base-like variant, T = 1.0	11	28

- With forcing, no pairwise difference between edits is significant. ERGORE vs MIT at T = 1.0: 29 vs 32, p = 0.38.
- [interpretation] Much of the measured difference between edits is about *getting the answer out*, not about *finding* it.

Doubt without abstention. [descriptive]

- Phrases such as "I'm not sure" or "let me guess" predict errors: with ERGORE at T = 1.0, only 3 of the 23 runs containing them are correct.

- Yet **49–85 % of wrong runs contain no doubt phrase at all.**
- **No run ever ends with an explicit abstention.** The probe, reading inside, classifies 96 % of wrong candidates as "not true" (4 % threshold).

3.5 Blunt pressure has a price

IFEval harm check. [pre-registered as a harm check] 120 identical prompts, same protocol:

	IFEval (strict, prompt level)
base	18/120
own direction, no closing push	74/120
ERGORE (with closing push)	**61/120** (vs 74: 11 : 24, p = 0.041)
WT-A (truth-gated trigger)	70/120 (vs 74: p = 0.54; vs ERGORE: p = 0.21; vs MIT: p = 0.005) [pre-registered reading: confirmed at ≥ 72, refuted at ≤ 64, so neither]
MIT edit	87/120

- Losses concentrate in keyword instructions (11/38 vs 22/38) and combined instructions.
- [interpretation] The unconditional push makes the model hurry and drop required details.

Empty commits. [descriptive] Of ERGORE's 26 errors on Set B (T = 0.6), 22 are runs that close their reasoning, often without any candidate, and then ramble or loop in the answer part, e.g. "I will formulate the response. Final.". Forced to answer at that point, only 4 of 22 are correct, often with a placeholder "1". The model genuinely did not have the solution.

3.6 WT: a truth-gated trigger in the weights [descriptive on the development set; fresh-problem test pre-registered, Section 3.9]

Construction. 16 rarely active neurons in layer 15 are rewritten:

weight	role
`gate`	truth direction above a high threshold (2 % of wrong candidates pass)
`up`	"a candidate just ended" detector
`down`	the **closing gradient** $\hat{g} = \partial \log p(\texttt{</think>})/\partial h_{15}$, averaged over correct candidate ends in practice runs

$\hat{g}$ is nearly orthogonal to the state-difference direction $e_\perp$ of our first gate. A push of size 5 along $\hat{g}$ raises $p(\texttt{</think>})$ from $\sim 10^{-12}$ to ~ 0.95.

Offline check on held-out practice prefixes, two bases (own direction only; ERGORE), thresholds 2 % and 4 %:

position	without WT	with WT
correct candidate end	$p(\texttt{</think>}) \approx 10^{-12}$, 0/8	**≈ 1.0, 6/8**
wrong candidate end	$\approx 10^{-10}$, 0/8	unchanged, 0/8
ordinary token	$\approx 10^{-12}$, 0/8	unchanged, 0/8

Real-generation test [pre-registered]: Set B, n = 60, two machines in blocks, T = 0.6. WT-A is WT built with the sign rule (Section 3.8) on top of our own loop direction.

criterion (fixed in advance)	result
K0 mechanics: no immediate or candidate-free early closes	met (0 and 0)
K1 no harm vs. the same model without WT	met: 31 vs 23 correct (only WT-A 9, only without WT 1; p = 0.021)
K2 closes ≤ 64 tokens after a correct candidate	**not assessable**: 2 opportunities (see below)
K3 token savings on problems both solve	met: mean 5,248 vs 14,776 (−64 %)
K4 truth probe intact on the new transcripts	**not met**: AUC 0.82 / 0.94 vs 0.99 / 0.996 (two blocks)

Following the reading fixed in advance, the gate to a confirmatory claim **stayed closed**. All later WT-A numbers are descriptive, except H-WT2 (Section 3.9).

- **Why K2 could not be measured.** WT-A closes right after result sentences (`So $\binom{10}{8} = 45$.`), *before* the model writes `\boxed{…}` or "the answer is". Our old candidate detector looked only for those.
- **The fix.** We wrote a broader detector (old candidates plus value statements "= X", "is X"). With it we fixed a new test (H-WT2) before any run on fresh problems existed.
- **A second seed** reproduces the development-set result: 33 vs 25 (+8 again). [descriptive]

Where the probe falls short. [descriptive] Correct candidates below the threshold are mostly tentative mentions during a self-check. In the 60 tokens before them, check words appear in 51 % of cases (vs 21 % above the threshold), hedges in 29 % (vs 17 %). [interpretation] The model has not yet *asserted* the answer at that moment. Lowering the threshold catches some of these but adds misfires (simulation: 2 % → 4 % threshold, +13 correct fires, +6 misfires).

Truth versus assertion (2 × 2). [descriptive] Does the probe read *truth* or just a confident *form*? The gate is conservative by design; this test asks *what* it reads, not *how often* it fires.

- **Setup:** On hard cases (correct candidates below the threshold, n = 38) we append one of four continuations: "So the answer is X." or "Let's check if it is X.", each with the true and with a wrong X. The probe is read at the end of X.
- **Wrong X:** a confident false assertion never crosses the threshold (**0 of 38**; upper 95 % bound ≈ 9 %).
- **True X:** a confident true assertion lifts **10 of 38 (26 %)** over the threshold.
- **Truth effect:** positive in 35 of 38 cases.
- [interpretation] The probe reads truth, not assertiveness. A clear statement of a true answer is a legitimate way to reach the bar without lowering it.

3.7 Negative results [pre-registered]

- **First gate (V2).** Same AND-structure as WT, but pushing along e⊥ and without ERGORE's closing push. It fails all three criteria:
  - Set B: 27 vs ERGORE 34 at T = 0.6, and 11 vs MIT 24 at T = 1.0;
  - empty commits 14 (target ≤ 6);
  - probe AUC 0.892 (target 0.93).
  - Without the closing push the model often never stops: 18 of 60 runs at T = 0.6, 44 of 60 at T = 1.0.
- **Tripling that push** changes nothing (28 vs 27).
- **Temperature.** With this model, T = 0.6 clearly beats the vendor-recommended T = 1.0 (ERGORE 34 vs 21 on Set B, p = 0.001).
- **Thinking switch** (decide before thinking whether to think at all).
  - **Hypothesis:** the inner signal at the end of the prompt (truth probe and candidate-end probe) predicts whether the model solves the problem *without* thinking, beyond difficulty level and prompt length.

  - **Result:** likelihood-ratio test p = 0.13, cross-validated AUC 0.72 → 0.73. **Refuted.**
- **ERGORE on fresh problems** (Section 3.9). "MIT level not reached" on 161–300.

3.8 A recipe: programmed reflexes in quiet neurons [method; transfer to other models untested]

The MIT edit *removes an obstacle*: it reverses one loop direction. WT goes one step further and *writes a conditional reflex* into the weights: **if a condition holds in a given context, push towards an action.** All three parts are measured from the model itself; nothing is trained.

The formula. For K rarely active SwiGLU neurons G in one mid-depth layer, with x the input of that layer's MLP:

> **reflex = effector × (condition ∧ context)**
>
> $\Delta h(x) = K \cdot \kappa_d \cdot \mathrm{SiLU}(\, \kappa_g \cdot (s_cond(x) - \tau)\,) \cdot \kappa_u \cdot (s_ctx(x) - \tau') \cdot \mathbf{e}$

part	what it is	in WT
condition s_cond	linear probe on x	"this value is true" (truth probe s_T)
context s_ctx	second linear probe on x	"a result just ended" (candidate-end probe s_C)
effector **e**	unit direction that triggers the action: the mean normalized gradient ∂ log p(action token)/∂h at the places where the action is wanted	"write `</think>`"
place	the K quietest neurons by activation RMS, outside other edits	16 neurons in layer 15 of 24

Writing it into the weights. SwiGLU already contains the multiplication, so the AND needs no runtime hook:

- `gate[G] = κ_g · (w_cond + (b_cond − τ) · v)`
- `up[G] = κ_u · (w_ctx + (b_ctx − τ') · v)`
- `down[:, G] = κ_d · e`

The vector v emulates the missing bias ($v^T x \approx 1$ on ordinary tokens).

Calibration rules, all fixed before testing:

- τ lets only a small share of wrong candidates through (2 %).
- κ_g brings the median correct candidate to a gate value z* = 3.
- **τ′ is the median of s_ctx on ordinary tokens.** This is the sign rule: SiLU is negative between about −4 and 0, so the context term must not be negative wherever the gate sits in that region. Otherwise the product turns positive and the reflex fires where it should not. Our first build (V2.1) broke this rule; see the leak below.
- κ_u scales a typical candidate end to 1.
- κ_d = κ / (K · SiLU(z*)) gives a total push of κ = 7 units along e at a typical firing position.

Mandatory offline checks. Measure p(action token) for the edited and the base model on identical texts:

position	must
correct candidate end	fire
wrong candidate end, ordinary token	stay at base level
last prompt token (first generated token)	stay at base level
early paragraph ends	stay at base level
sentence end after a refuted check	stay at base level
intermediate result (≠ final answer)	stay at base level
answer part (after the action)	stay at base level

The first build passed the first three rows on thinking positions but leaked at the prompt end: 9–10 of 30 runs closed at once (Section 3.6). That is why the list has seven rows. WT-A passes all thinking-side rows; it still fires in the answer part (second `</think>`, see the lesson below).

Evaluation protocol:

- block design across machines;
- criteria fixed in advance (mechanics, no harm, effect, savings, probe intact);
- confirmation only on untouched problems;
- answer forcing and an instruction-following check (IFEval) as side conditions.

Transfer to another SwiGLU model, in six steps:

1. Train the two probes on the model's own residual stream at a mid-depth MLP input.
2. Measure the effector gradient at the desired action points.
3. Select the quiet neurons.
4. Write the three weight blocks.
5. Pass all mandatory checks, including the sign rule.
6. Evaluate with a reading fixed in advance.

All scripts are in the released code. Whether the recipe carries over to larger models is the question of Phase 2.

Lesson from WT-A [descriptive]: the effector must fit the *phase*, and it must leave a *state*, not just a token.

- After a trigger close, WT-A reopens its reasoning in the answer part (a second `</think>`) in 83–100 % of runs, even when the value had been stated six or more times before. This produced 46 % of all tokens on the development set.
  - On fresh problems (161–300) the figure is 77 % of all closes and 58 of 61 correct runs; MIT reopens after 16 % of its closes.
  - WT-A's correct answers almost always come after its *own* second close.
- [interpretation] The gradient effector $\hat{g}$ raises `</think>` for exactly one step. One token later the candidate-end context is gone and the 16 neurons fall silent. The attention layers then see a reasoning that was cut off mid-check, and the model treats the close as foreign.
- A fix therefore needs an effector with *depth of effect*: a push that starts the model's own way of finishing instead of forcing a single token. One candidate is the part of MIT's direction that our own does not explain ($r_\perp$). Its top tokens are natural ending words ("thereby", "accordingly", end-of-text), and it is orthogonal to $\hat{g}$ (cosine 0.005).
- The pre-test of depth of effect (Section 3.9) found no such effector yet. A single push along $\hat{g}$ or $r_\perp$ acts only while it is applied, and spreading it over three steps leads to *more* renewed thinking. The reviewers'

proposals therefore aim at accompanying the model's own closing phrase (Section 6).

A second instance (proposal). The same formula can express an honesty reflex:

- condition: "the truth probe stays low";
- context: "about to commit";
- effector: towards an abstention phrase ("I don't know").

Honesty step 1 tests only the condition, as an external rule. [pre-registered]

- **Setup:** The frozen layer-13 truth probe is read at the *delivered* answer. If it stays below its frozen threshold, the answer is replaced by "I don't know". Runs without an answer are first forced to answer.
- **Data:** 1,280 runs from all sets and models.

	correct	wrong	abstained	score S_2 = (correct − 2·wrong)/N
official answers	592	–	–	+0.20
forced answers	726	552	–	−0.30
forced + probe abstention	673	225	380	**+0.17**

- **Both criteria met:** AUC on forced answers 0.83 (required ≥ 0.75), and the harness beats forcing on S_2.
- 327 of the 380 abstentions would have been wrong; 53 correct answers were lost.
- **Risk–coverage:** answering only the 50 % most confident gives 87 % accuracy, the 30 % most confident give 95 %.
- On the fresh problems alone: AUC 0.83 (forced answers 0.80), S_2 +0.13 with the harness vs −0.24 forced.
- [interpretation] The inner truth signal does not only trigger commitment; it also tells when *not* to commit. Writing the abstention itself into the weights is the next step.

3.9 Fresh problems: MATH-500 161–300

> *"A teacher does not give the candidate a push from outside either."* — D. U., 8 Nov 2026

Design.

- Problems 161–300 were untouched until this step.
- The laptop ran 161–260, the second machine 261–300. Every pairwise comparison runs on the same machine.
- Same protocol for all models, seed 0; a second seed for parts of the data.

Main verdict [pre-registered]. ERGORE does not reach MIT level: 65 vs 80 of 140 (7 : 22, p = 0.008).

block	ERGORE	MIT	note
161–260 (laptop)	51	58	seed 0
261–300 (second machine), seed 0	14	22	3 : 11
261–300, seed 1 [descriptive]	16	16	6 : 6

[interpretation] The development-range tie did not transfer. On one block, MIT's own seed-to-seed variation (22 → 16) is as large as the gap. The gap is real, but probably closer to 5–7 points than to 11.

WT-A on fresh problems.

	WT-A seed 0	WT-A seed 1 (161–260 only)	MIT
correct [descriptive]	61/140 (161–260: 44/100)	47/100	80/140 (161–260: 58/100)
H-WT2: closes ≤ 64 tokens after the correct value, per opportunity [pre-registered]	**65 % (55/85)**	54 %	36 % (37/102), p = 0.0001
H-WT2: fast closes that are right [pre-registered]	**61 % (55/90)**	57 %	44 % (37/85)
tokens on problems both solve	7,146	–	13,233

Inside view: a teacher-forced trigger audit [descriptive; readings fixed in advance]. We replayed all 140 WT-A transcripts with and without the 16 neurons, reading the gate, the context term, the push and the sampled close probability at every position.

- **The trigger carries WT-A's successes.**
  - 69 of 102 first closes are trigger-driven: the sampled close probability is < 2 % without the neurons and ≥ 20 % with them.
  - 46 of these 69 follow the correct value, 18 a wrong one, 5 elsewhere. They yield **51 of WT-A's 61 correct answers**.
  - WT-A's natural closes (25) are mostly at wrong values; only 5 are correct.
- **Truth recognition is partial.** The gate opens at 17 % of correct and 6 % of wrong value statements (AUC of the gate value 0.68). Mostly it stays just above threshold, too weak for a full trigger.
- **The counter-push of the negative SiLU lobe is small:** 1.7 closes lost across 140 runs.
- **A relative threshold does not help** [exploratory]. A within-run z-score of the gate value separates worse (AUC 0.61).

Where the gap comes from [descriptive, post hoc]:

	WT-A	MIT
correct value appears in the reasoning	85	102
… closes right at it	55 → 48 delivered correctly	37 → 33
… does not close right at it	30 → **7** correct	65 → **44** correct
tokens from first correct value to close (median)	1,018	12,289

- **When the trigger fires:** on early, clearly stated solutions of easier problems.
  - First correct value at a median of 719 vs 1,963 tokens.
  - Levels 1–3: 75 % of these runs close at the value; levels 4–5: 50 %.
- **Silence is often a delivery failure.** 32 WT-A runs close the reasoning without a `\boxed` answer. In 14 of them, the correct value appears early in the answer part and is then lost.
  - How often does WT-A reopen its reasoning after a close (a second `</think>`)? It depends on which runs are counted:
    - in 83–100 % of the development runs;
    - in 77 % of all its closes on these fresh problems (58 of its 61 correct runs);
    - in 12 of these 32 silent runs.
  - MIT reopens after 16 % of its closes.
- **The truth signal fades with loops and difficulty, not with depth.** We fit a logistic regression on the correct ones among 109,561 value statements from WT-A and MIT transcripts (same probe, same layer-15 input), standardised, with a run-level bootstrap.

factor	coefficient for "gate open" at correct values	95 % interval
depth (log tokens)	**+0.26**	+0.01 to +0.49
local repetition	**−0.80**	−1.08 to −0.50
difficulty level	**−0.68**	−1.06 to −0.32

  - Nested models without the bootstrap show the same reversal. Depth alone gives −0.49, adding difficulty −0.27, adding loops +0.46, adding the transcript source +0.26. The last step is the full model of the table.
  - Without loops, the gate still opens at ~60 % of correct statements at 8,000–16,000 tokens. Inside loops beyond 16,000 tokens it opens at 15 %.
  - [interpretation] The probe reads the state of the reasoning, not the distance from its start. A depth-dependent threshold is not indicated.
- **The trigger would not help MIT.** [estimate, not a measurement] We replayed MIT's transcripts with the trigger neurons added.
  - The trigger would cut MIT's long check short, often at a wrong value.
  - The estimate stays below MIT's actual score under all threshold settings. Waiting for repeated triggers on the same value comes close, but does not exceed it.
  - The estimate depends on assumptions about delivery after an early close, so we report it only as a reason for not running this combination.
  - [interpretation] A truth-gated trigger pays off where a model has no good way to finish on its own (WT-A vs its base: +8 in both seeds). It does not pay off where the model already checks and commits well.
- **Depth of effect** [pre-test, readings fixed in advance; 30 trigger closes at correct values, greedy decoding].

push at the trigger point	closes	p(`</think>`) when closing is blocked, steps 0 / 1 / 2 / 5	closing sentence before	re-thinking after
WT-A's real trigger	26/30	0.75 / 0.27 / 0.03 / 0	0	4
none (base edit)	0/30	0	–	–
$\hat{g}$ once ($\kappa = 7$)	30/30	1.00 / 0.02 / 0 / 0	0	4
$\pm r_\perp$ once ($\kappa = 7$)	0/30	0	–	–
$\hat{g}$ spread over 3 steps	17/30	0.01 / 0.36 / 0.32 / 0	0	11

  - Every push acts exactly as long as it is applied; no lasting "done" state forms. No variant produces a closing sentence.
  - A single push along $r_\perp$ does nothing. MIT's closing phrases therefore come from an edit that acts continuously, not from a kick.
  - [interpretation] An inner "done" has to build up along the closing stretch. The next candidate fires *while* the model writes its own closing phrase, and only when the truth signal is high. At exactly those points the probe separates right from wrong values with AUC 0.80–0.85.

3.10 Answer mode ("INT")

> *"That is the first pair of shoes. Let us call it INT, for intuition."* — D. U., 8 Nov 2026

Qwen3.5's official non-thinking template closes the thinking block before generation starts. The model then reasons visibly in its answer, with an obligation to deliver and no separate place for unbounded checking. Ma et al. (2025) report that skipping the thinking block can be competitive under token budgets. Here it happens in an edited 0.8B model at MIT's full protocol.

- **Development range** [descriptive]: our own-direction model, without thinking block, same protocol and budget.
  - MATH-500 1–160: **77 of 160**, exactly as many as MIT with thinking (22 : 22). It uses ~8,200 instead of ~21,000 tokens.
  - Set B: 32 vs 23 for the same model with thinking (14 : 5, p = 0.064).
  - All 109 answer-mode hits end regularly (end-of-turn). None comes from a loop or budget cut-off, so this is not a grading artefact.
- **Is answer mode only good for easy problems?** [descriptive, development data] No.
  - By official difficulty (1–160): answer mode 26 vs MIT-with-thinking 22 on levels 4–5, and 33 vs 37 on levels 1–2.
  - Same model in both modes (Set B, two seeds for thinking): on problems the model answers quickly (< 1,000 tokens), answer mode gets 21 and thinking 19 and 19 of 26.
  - On problems where the answer takes long (> 8,000 tokens), answer mode gets 8, thinking 2 and 3 of 25.
  - The author's image for this is a house. The answer mode is the *front room*; the thinking block is the *back room*; between them lies a *hallway* whose exits lead to "closing time" (deliver and stop) or to the "break room" (pause, abstain, retry).
  - [interpretation] For this 0.8B model, the back room is not where hard problems get solved; it is where the model gets lost.
  - Answer mode also shows its own speed signal: answers within 1,000 tokens are right ~75 % of the time, answers beyond 8,000 tokens ~25 %.
- **Fresh problems 161–300** [pre-registered: holds if ≥ MIT − 7 of 140, refuted if ≤ MIT − 14]: answer mode **71**, MIT with thinking **80**. The result is **open**.
  - On 161–260, 57 vs 58; on 261–300, 14 vs 22.
  - All answer-mode runs end regularly. Per correct answer they need 14,800 instead of 36,600 tokens (161–260).
- **MIT in the front room** [pre-registered]: MIT's edit without thinking block reaches **59 of 140** (6 : 27 against MIT with thinking, p = 0.0003). The reading "the back room pays off for MIT" holds.
  - Our front room (71) beats MIT's front room (14 : 26, p = 0.08).
  - [interpretation] A back room is worth entering only when it has a working exit. MIT's edit builds that exit; ours builds a stronger front room.
- **Checks** [pre-registered]:
  - Presence penalty: the answer-mode advantage on Set B 1–20 is +2 both with penalty 0 and with 1.5. No artefact.
  - Day-to-day stability: with a second seed on 261–300, answer mode goes from 14 to 16, but only 75 % of problems are solved the same way. MIT with thinking also reaches 75 %. The author's guess that the front room is more stable is refuted.
- **Cascade "answer mode → truth probe → thinking run → I don't know"** [pre-registered; precondition not met, so descriptive].
  - **Precondition:** the frozen truth probe must separate answer-mode answers on 161–300 with AUC ≥ 0.75. It reaches 0.68 (0.76 on 1–160). The cascade is therefore reported only descriptively.

  - **Main variant:** thresholds calibrated on 1–160, MIT as fallback. On 161–300 it reaches S_2 +0.32 against +0.27 for MIT alone, but with only 67 correct answers (MIT: 80). The reading required $S_2 \geq +0.25$ *and* at least 75 correct answers, and it counted as refuted at $S_2 \leq +0.19$ *or* fewer than 75 correct. It is therefore **refuted** despite the higher S_2.
  - **Cost:** tokens rise rather than fall, because most problems end up in the thinking branch.
  - **Reason:** in answer mode the probe is shifted far upwards. Correct answers have a median score of +3.4, wrong ones +1.9 to +2.6, against a frozen threshold of −1.2.
  - [interpretation] The probe was trained on thinking text. In the front room every statement sounds assertive. A front-room cascade needs a probe trained on answer text.
- **Second chance ("dawn")** [pre-registered]. The 69 WT-A runs that delivered nothing were retried in answer mode with notes. The notes were the sentences around the model's own strongest truth flashes, chosen without access to the solution. The control was answer mode without notes on the same problems.
  - Notes 22, control 25 (p = 0.66). The effect of the notes is **refuted**.
  - [descriptive] The second chance in the front room itself rescues a third of the silent runs; WT-A had delivered nothing on them.
  - Notes help only when they contain the correct value (8 vs 5 of 10). The strongest truth flashes point to the correct value in only 10 of 65 runs.
- **Crossing out** [descriptive, rough marker]. In the back room, correct values are retracted ("wait", "let me check", …) *more* often than wrong ones: WT-A 74 % vs 60 %, MIT 59 % vs 45 %. The truth signal is the same whether a value is retracted or kept. In the front room, retraction is rare (18 % vs 14 %).
  - [interpretation] Retraction in the back room is a habit, not a judgement about truth.
- **Signposts** [descriptive]. Before MIT's natural closes the model writes its own closing phrases ("answer" 31 %, "final" 14 %, "ready" 8 % in the last 12 tokens). Before WT-A's trigger closes these are mostly missing.
- **Quick errors are not blind** [descriptive; corrects an earlier reading]. Among quick answer-mode answers (< 1,000 tokens), none of our *earlier* signals separates right from wrong: retractions, doubt words, distinct values and the thinking-text probe give AUC 0.48–0.54. We first called these "confident errors". A probe trained on answer text does separate them (next item).

> *"Several intuitions confer in the front room. If they agree, it is closing time. Only the disputed cases go to the back room."* — image agreed with D. U., 8 Nov 2026

- **Asking the intuition twice** [pre-registered]. The same model in answer mode, seeds 0 and 1. Rule: "same answer" when both runs deliver equivalent answers (checked as in grading).
  - **161–260** (100 fresh problems): same answer on 38 problems, **37 right (97 %)**; different answers on 62, seed 0 right in 32 %. The criterion (difference ≥ 30 points and ≥ 70 % when agreeing) holds with +65 points. Fisher one-sided $p \approx 9 \cdot 10^{-12}$.
  - **40 new training problems** (never used before): same answer on 17, 13 right (76 %); different on 23, 13 % right. The criterion holds with +63 points, but at clearly lower precision.
  - **Untouched problems 301–370**: same answer on 31, **28 right (90 %)**; different on 39, 18 % right. The criterion holds with +72 points.
  - [descriptive, pooled] Over the three confirmations, 78 of 86 agreeing answers are right (91 %). The 97 % of the first test is the upper end. Each set was pre-registered on its own; the pooled figure was not.

  - [interpretation] This is self-consistency (Wang et al., 2022) used at the point of commitment. Agreement does not vote on the answer; it decides *whether* to deliver.
  - [descriptive] On the already seen block 261–300 the same pattern appears: 10 of 12 vs 4 of 28.
- **Asking three times** [pre-registered; 161–260, seeds 0, 1, 2].
  - A majority of three (at least two equivalent answers) delivers on **62** problems, 55 of them right (89 %).
  - The reading required ≥ 48 deliveries and ≥ 90 % precision, so it is **open** by one point.
  - The second reading, that the net score (correct − 2·wrong)/N must not fall below agreement of two, holds: +0.41 vs +0.35.
  - All three agree on 29 problems, and all 29 are right.
  - [descriptive] The other seed pairs repeat the pattern: 0/2 agree on 42 problems (40 right), 1/2 on 40 (35 right).
- **A probe for the front room** [pre-registered; result "open"]. Same recipe as the truth probe, but trained on answer-mode texts: 239 delivered answers from Set B and 1–160. It reads the state at the last token of the delivered `\boxed{}`.
  - Layer and regularisation were chosen by grouped cross-validation on training data only (layer 13, CV AUC 0.81), then frozen.
  - Test on 161–300: AUC **0.763**. Answer length and difficulty alone reach 0.723, and the combination adds only +0.04. The reading fixed in advance required +0.05, so the result is **open**.
  - The thinking-text probe reaches 0.681 at the same positions.
  - [descriptive, post-hoc subsets] On quick answers (≤ 2,000 tokens) the new probe reaches 0.794, against 0.63 for length and difficulty and 0.63 for the old probe. Delivering only quick answers above the probe's median gives 26 of 29 right.
  - It transfers to MIT's answer-mode texts (0.776) and to new samples of the same problems (0.836).
  - **Confirmations** [pre-registered, frozen probe and frozen baseline]:

set	answers	probe AUC	length + difficulty	gain	quick answers (≤ 2,000): probe vs baseline
40 new training problems	79	0.852	0.841	+0.012	0.823 vs 0.736
untouched 301–370	138	0.765	0.717	+0.048	0.606 vs 0.599

    - Both main readings are **open**: the probe reads correctness well, but answer length and difficulty already carry most of that information.
    - The quick-answer advantage from the post-hoc subset **did not replicate** on the untouched problems. It was fixed in advance as a second reading, and that reading is refuted.
  - **Added value without harm** (author's criterion: "only if it adds value and does no harm, measurably") [pre-registered, 301–370]:
    - *As referee*: when two intuitions disagree and both deliver an answer, take the one with the higher probe score. On 37 such problems, one of the two answers was right in 18. The probe picks 11 of them; the first run alone has 7, the shorter answer 7. Reading: **added value**.
      - It is a fragile sample: only 18 problems decide, and the paired bootstrap 95 % interval for the gain is −1 to +10 correct answers.
      - [descriptive] Under "always answer", the referee raises the count from 35 to 39 of 70. The two single runs score 35 and 39, so the mean of a single run is 37.

        - *As an extra delivery gate*: the score falls from +0.31 (agreement alone) to +0.13. Reading: **harm**. The probe's extra deliveries are right in only 3 of 11 cases.
        - [interpretation] Agreement decides *whether* to deliver; the probe decides *which* answer when the intuitions disagree.
- **Switching rooms** [pre-registered]. Deliver when two intuitions agree, otherwise use a thinking run.
    - With WT-A's back room as fallback: 54 of 100, against 57 for answer mode alone. Refuted.
    - [descriptive] With MIT's back room: 63, against 57 and 58 for each alone.
    - [descriptive] With three intuitions (majority, otherwise back room): WT-A's back room as fallback gives **62**, against 57 for answer mode alone and 44 for WT-A alone. MIT's back room gives 66, against 58 for MIT alone. With three intuitions our own back room adds value for the first time.
    - [descriptive, 161–300] "Agreement alone" as the rule for delivering, otherwise "I don't know": 50 of 140 delivered, 94 % right, score (correct − 2·wrong)/N of +0.29 against −0.48 for always answering.
    - [descriptive, 161–300] The gain of "agree → answer mode, otherwise MIT" (84 vs 80) comes entirely from the agreeing set (47 vs 43 there). The rule is a selector, not an amplifier, and two full answer-mode runs make it costlier in tokens (54,000 vs 37,700 per correct answer).
    - [pre-registered, 301–370] With the probe as an extra gate, the coupled rule beats a length-only rule only by +0.04 in score, at 74 % precision (required: +0.05 and 80 %). The result is **open**.
    - **"Agree → answer mode, otherwise MIT's back room" on the untouched problems 301–370** [pre-registered]. The rule holds if it reaches at least MIT alone, with ≥ 90 % precision on the agreeing set and fewer tokens per correct answer than MIT.
        - Result: **40 of 70**, against 38 for MIT alone and 35 for answer mode alone. Precision on the agreeing set is 90 % (28 of 31).
        - The token criterion fails: 56,800 vs 41,000 per correct answer. The reading is therefore **open**.
        - Decomposition: on the 31 agreeing problems, answer mode has 28 and MIT 26. On the other 39, MIT has 12 and answer mode 7. The gain comes from both rooms: precision in front, MIT's exit at the back.
        - [descriptive] On these problems answer mode alone (35) stays close to MIT with thinking (38), at 17,000 vs 41,000 tokens per correct answer.
- **Closing at one's own signpost** [WT-A pre-registered, MIT descriptive]. We cut reasoning texts right after a value that follows the model's own closing phrase ("the answer is …"), inserted `</think>` and regenerated the answer part. The control was a value without such a phrase in the same run.
    - MIT (74 runs, 1,024-token cap): it delivers in 36 % of cases at its signpost and in 18 % at the control (17 : 3, p ≈ 0.003). All 19 deliveries after a correct value are right.
    - WT-A (19 runs) first gave 0 and 0, but with a 1,024-token cap that was too short for WT-A. Its natural correct answer parts have a median of ~2,000 tokens.
    - In the fair re-test with 4,096 tokens [pre-registered]:
        - WT-A delivers in 32 % at its signpost and in 21 % at the control (3 : 1). The result is open.
        - Where its own signpost *and* the truth gate coincide, it delivers in 5 of 9 cases, all of them right.
        - Over all runs, a reflex built on this would not gain anything: 59 instead of 61. It also cuts runs that would later have delivered correctly on their own.
    - The same re-test confirms the replay of WT-A's 32 silent answer parts: even with 4,096 tokens, only 2 of 32 deliver.
    - [interpretation] The model's own signposts, together with the truth signal, mark the moments where a close holds. An inserted `</think>` still does not replace an inner "finished".

4. A lens: where do a model's failures live?

The "fox den" is the author's image for this lens. We sort every problem into five "chambers":

- **S**: solved reliably;
- **G**: grey zone, sometimes solved;
- **E**: decision reserve, the solution appears but the model does not commit;
- **F**: false commit, it commits stably to a wrong reading;
- **K**: no trace, the solution never appears.

Each chamber needs a different lever:

chamber	lever
E	a commit trigger (WT)
F	a second look
K	an honest "I don't know", handed to tools or a larger model

[interpretation] Our data suggest that weight edits on a 0.8B model mainly work in E. Pushing in K produces confident wrong answers rather than solutions.

5. Limitations

- One model (0.8B). Mostly one seed; second seeds cover Set B and parts of 161–300.
- Set B and MATH-500 1–160 were reused for development.
  - The step to fresh problems showed exactly the expected optimism: ERGORE's tie did not transfer.
  - WT-A's lead over MIT on Set B (31 vs 27) reversed on fresh problems.
- Seed variation is large relative to some gaps; MIT itself varies by 6 of 40 between seeds on one block.
- Several mechanism results are post hoc and rest on small cells: the 14 lost answers, the threshold shifts, the signpost replays. They generate hypotheses; they are not established effects. (The re-thinking rate after a trigger close, first seen on the development set, has since been measured on all fresh closes as well.)
- The trigger audit replays sampled transcripts with teacher forcing in transformers (bf16); the runs themselves used llama.cpp (q8_0).
- IFEval and MATH are narrow proxies for "instruction following" and "reasoning".
- Replay tests need an answer cap that fits the model. We first used 1,024 tokens for WT-A, whose own correct answer parts have a median of ~2,000 tokens. We report the fair re-test with 4,096 tokens; the first verdicts are kept in the log.
- The front-room probe rests on 239 training answers. Its strongest numbers (quick answers) came from subsets chosen after the fact and did not replicate on untouched problems.
- The front-room confirmations are small: 17 to 38 agreeing answers per set. Pooled precision (91 %) is more trustworthy than any single set (76–97 %).
- "Agreement" is checked with the grading tool (math-verify). Two equivalent but differently written answers count as agreeing; two wrong answers that happen to match count as a confident error.

6. Next steps

- **Running while this report is published (public pre-registration).** We test the council of intuitions with *our own* back room on 150 fresh MATH training problems that were never used. They have the same level mix as MATH-500.
  - Runs: answer mode with seeds 0, 1 and 2, plus WT-A and MIT with thinking.
  - Rule: if at least two of the three intuitions agree, deliver that answer; otherwise take WT-A's thinking answer.
  - Readings, fixed before any of these runs existed:
    - (H1) the rule "holds" if it solves ≥ 5 more problems than answer mode alone, and is "refuted" at ≤ +1;
    - (H2) against MIT: "beats MIT" at ≥ +5 with one-sided McNemar $p < 0.05$, "at least level" at ≥ 0, "below MIT" at ≤ −5;
    - (H3) agreement of two intuitions "holds" at ≥ 85 % precision and is "refuted" below 75 %;
    - (H4) a majority of three "carries" at ≥ 85 % precision with ≥ 10 points more coverage than agreement of two.
  - The result will be reported in the next version, whatever it is.
- **Inner commitment instead of a forced token.** Build an effector with depth of effect. Single pushes along $\hat{g}$ or $r_\perp$ failed (Section 3.9). The remaining candidates are a continuous, gated version of MIT's exit direction and support for the model's own closing phrase (both below). Accept a candidate only if it produces a closing sentence and a clean answer instead of renewed thinking.
- **Front room first, back room with an exit.** This is the author's coupled design, now measured in parts:
  1. Several intuitions answer in the front room.
  2. Deliver when they agree. The probe referees disputes.
  3. Only disputed cases switch to the back room.
  4. That switch pays only once our back room has an exit like MIT's.
- **Building the exit from the inside.** All three external reviewers independently proposed the same next build: MIT's residual "exit" direction, gated so that it acts only inside the thinking block, while our own edit keeps the front room. A cheap teacher-forcing pre-test decides whether to build it.
- **Raising the model's own signpost instead of `</think>`.** Push the model towards its own closing phrase ("So the final answer is …") when the truth gate is high, and leave the actual close to the model.
  - Pre-test: 16-token continuations at confirmed positions against positions with unfinished intermediate values.
- **A one-sided switch.** Distil the agreement of several intuitions into a one-pass head. Then let it only *extend* checking when unsure, never cut a confident answer.
- **Threshold with a guarantee.** Calibrate the trigger threshold with distribution-free risk control (Learn-then-Test) on run-level maxima.
- **Honest abstention.** *"'I don't know', and that is being honest!"* — D. U., 6 Nov 2026
  - Start with an external harness: a run without any "true" flash ends in "I don't know".
  - Then try local abstention directions at `</think>`, built with the recipe of Section 3.8.
  - Beyond mathematics: a planned "honesty battery" with everyday questions. What does the inner signal show where nothing can be checked?
- **Larger models.** Transfer to a 35B mixture-of-experts model.

7. Reproducibility and licenses

The release accompanying this report, at github.com/NeoCortex0815/knowing-is-not-committing, contains:

- the code as used: evaluation client, edit builders, the truth probe and every test and verdict script, numbered in the order they were written;
- the dated pre-registration and analysis log (German), with machine names, paths and details of the author's private systems redacted;
- one table per evaluation run (problem id, extracted answer, correct, end reason, tokens) and the outputs of the verdict scripts;
- the ids of the MATH training problems we used.

Not included are model weights (they can be rebuilt with the code), the datasets (obtain them from their sources) and the full transcripts (large; available on request). Our code is licensed under Apache-2.0. Qwen3.5-0.8B is Apache-2.0; the MIT logs, IFEval and MATH stay under their own licenses and are only cited.

8. Acknowledgments

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Appendix A. Images that shaped this work

The author thinks in images. We keep the ones that became testable questions. Each is given in English, followed by the German original, by what it led to and by what came out. All other images and questions, with date and time, are in the pre-registration log.

1. *"Find out whether the model has the solution, and when, and let the inner push through, despite the pressure."* („Schauen, ob das Modell die Lösung hat, wann – und den inneren Schubs zulassen, trotz Druck.", 6 Nov)
   - This led to the three questions of the project: does the model hold the solution (truth probe), when does it recognise it (trigger audit), and can an inner push let it commit (WT reflex)?
   - Outcome: knowing, yes; committing remains the bottleneck.
2. *"Is the model simply unable to say 'I don't know'? … 'I don't know', and that is being honest!"* („Ist das Modell per se nicht fähig zu sagen ‚ich weiß es nicht'?", 7 Nov; „Weiß ich nicht – bin ich ehrlich!", 6 Nov)
   - This led to answer forcing and the honesty step.
   - Outcome: abstaining when the inner truth signal stays low turns forced answering from harmful into useful. The score (correct − 2·wrong)/N goes from −0.30 to +0.17 [pre-registered].
3. *"Was there a twitch? If so, was it because the solution is true or not? How strong was it?"* („Gab es ein Zucken? Wenn ja, war es, weil die Lösung wahr ist oder nicht? Wie stark war es?", 8 Nov)
   - This led to the teacher-forced audit of WT-A's trigger on fresh problems (Section 3.9).
4. *"A teacher does not give the candidate a push from outside either."* („Ein Prüfer gibt dem Prüfling auch keinen Schubs von außen.", 8 Nov)
   - This led to the principle of inner commitment.
   - Outcome: every external push we tried failed to produce an inner "finished": the gradient push, the residual direction, and an inserted `</think>`, even at the model's own closing phrase.
5. *"That is the first pair of shoes. Let us call it INT, for intuition. … The front room, the back room, and a hallway between them, whose exits lead to closing time or to the break room."* („Das ist das erste Paar Schuhe. Nennen wir es INT für Intuition." – „Vorderzimmer … Hinterzimmer … Flur liegt dazwischen. Flur könnte Ausgabe sein zum Feierabend oder zum Pausenraum.", 8 Nov)
   - This led to answer mode as a path of its own and to the house model of Section 3.10.
   - Outcome: our strongest own path, 71 of 140, at less than half the tokens per correct answer.
6. *"The signposts are put up by whoever lives in the house."* („Die Schilder hängt derjenige auf, der das Haus besitzt bzw. darin wohnt.", 8 Nov)
   - This led to the signpost tests.
   - Outcome: at its own closing phrase, MIT delivers twice as often as without one (36 % vs 18 %). Where WT-A's own signpost and its truth gate coincide, it delivers in 5 of 9 cases, all of them right.
7. *"Several intuitions confer in the front room. If they agree, it is closing time. Only the disputed cases go to the back room."* (image formulated in the working session and adopted by the author, 8 Nov)
   - This led to the agreement tests.
   - Outcome: agreeing intuitions are right 91 % of the time over three confirmations. A majority of three delivers more often, at 89 %, and the front-room probe referees disputes.